CURRICULUM SCHEME & SYLLABUS: SE SEMESTER IV

1. Course Structure and Subjects Covered in Semester IV

Which subjects are covered in the Semester IV syllabus? Under the University of Mumbai NEP 2020 Scheme, the subjects covered in the Semester IV syllabus for Second Year Electronics & Computer Science (ECS) are as follows:

Code	Subject / Course Name	Scheme (L+P+T)	Credits
ECC401	Applied Mathematics IV (Calculus & Transforms)	3L + 1T	4
ECC402	Microcontroller Applications & Embedded Systems	3L + 2P	4
ECC403	Database Management Systems (DBMS)	3L + 2P	4
ECC404	Principles of Communication Engineering	3L + 2P	4
ECC405	Signals and Systems / Signal Processing	3L + 0P	3
ECM401	Mini Project 1B (Embedded Software System)	0L + 4P	2

Total semester academic credits: 21 Credits. Practical laboratory sessions are held in dedicated department labs with modern simulation and hardware kits.

2. Detailed Module Outline for Core Subjects

- **Microcontroller Applications:** Instruction sets and architecture of ARM Cortex-M and 8051 controllers, timer and interrupt handling, GPIO interfacing, serial protocols (UART, SPI, I2C), PWM generation, and embedded C firmware design.
- **Database Management Systems (DBMS):** Relational database concepts, ER diagrams, SQL DDL/DML, functional dependencies, normalization (1NF, 2NF, 3NF, BCNF), indexing (B-trees, hashing), transaction processing, ACID properties, concurrency control, and introductory NoSQL databases.
- **Signals and Systems:** Continuous and discrete-time signals, Linear Time-Invariant (LTI) systems, Fourier transform, Laplace transform, Z-transform, frequency response, and digital filter fundamentals.

3. Examination Evaluation and Credit Award Pattern

- **Internal Assessment (IA):** Two internal unit tests of 20 marks each; the average score of both tests is incorporated into final grade calculation.
- **End Semester Examination (ESE):** University theory paper of 80 marks with a duration of 3 hours covering all six syllabus modules.
- **Term Work (TW) & Oral Exam:** 25 marks term work based on journal submission and mini project defense evaluated jointly by external and internal examiners.